



BRIEF REPORT | ANTHROPOLOGY |



Radiocarbon dating of Jerusalem's Siloam Dam links climate data and major waterworks

Johanna Regev , Nahshon Szanton, Filip Vukosavović , and Elisabetta Boaretto [Authors Info & Affiliations](#)

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Abstract

Using well-established microarchaeological sampling methods, we reached a precise radiocarbon date of 800 BC for the Siloam Pool's monumental water dam in Jerusalem. This date is a critical link connecting several imposing waterworks constructed at that time. Climate data pointing to droughts and flash floods during the last decades of the 9th century BC provide a logical framework for the reasons behind such endeavors. These included the fortification of the city's primary water source, the Gihon Spring, and the redirection of the water into the city through a channel to an artificial reservoir created by building the Siloam Dam at the end of the Tyropoeon Valley, which blocked



the drainage of rain and redirected spring waters.

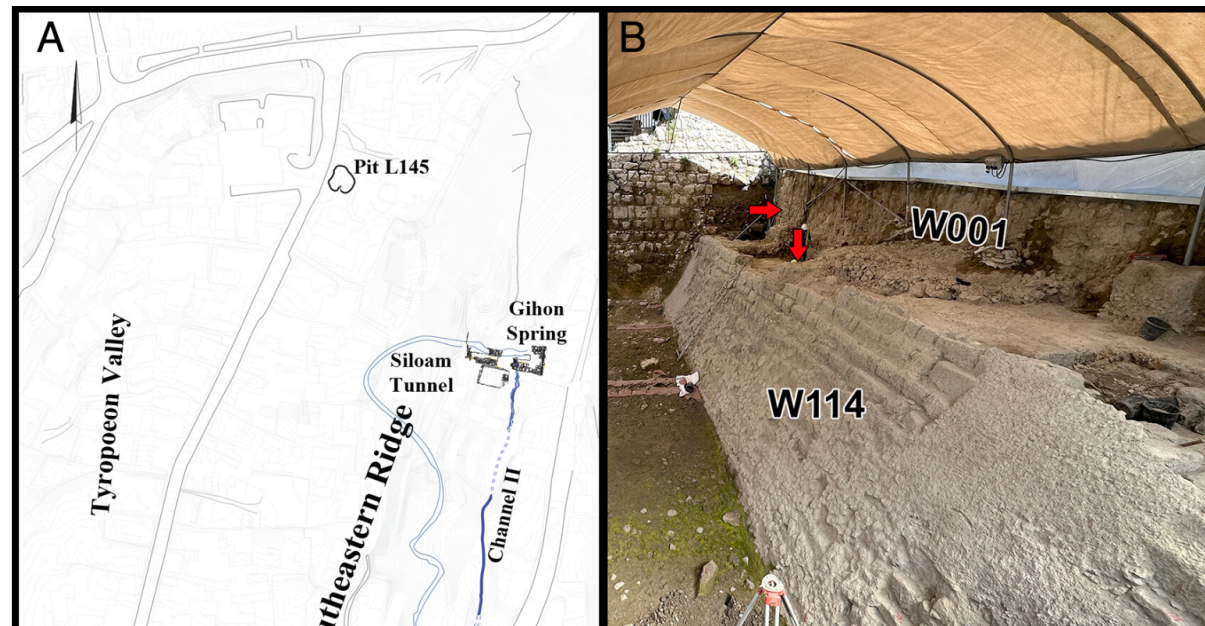
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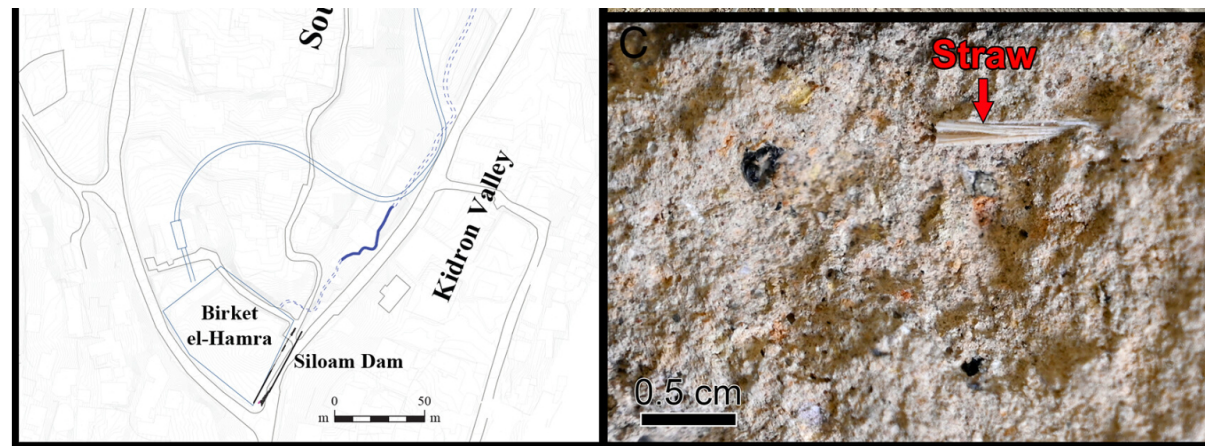
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The need for water has always been a key factor for the placement of settlements in the Levant. Thus, it is no surprise that Jerusalem's earliest settlement was built around the Gihon Spring, located at the eastern slope of the Southeastern Ridge, popularly known as the "City of David" (Fig. 1) (1). Although several ancient water system elements have been exposed, the lack of absolute dating and vastly different attributed dates have made it impossible to consider them as a single system (2). With the new radiocarbon dates, a picture of a well-planned and executed large water system is emerging.

Fig. 1.





(A) The study sites. Graphics by Manuel Cimadevilla, Israel Antiquities Authority. (B) The Siloam Dam excavation. Arrows mark the sampling locations. (C) A representative straw fragment embedded in the mortar.

The large tower built of megalithic stones around the water source of the Gihon Spring was dated by the excavators to the Middle Bronze Age (MBA), c.1800 BC (2). This dating was challenged by a sequence of radiocarbon measurements taken underneath the lowest course of stones, with three 10th to 9th-century dates. The latest date ranged between 830 and 800 BC (3). This tower encompasses entrances to two water channels leading down to the southernmost part of the Southeastern Ridge: Channel II and the Siloam Tunnel.

Like the tower, the northern part of Channel II was dated to the MBA (2). Channel II went out of use following the carving of the Siloam Tunnel, whose level is 2.5 m lower.

Two radiocarbon dates were previously measured from the lowest plaster layer above the bedrock of the Siloam Tunnel's floor from two locations. A wood sample was dated 810 to 780 BC, and a short-lived plant was calibrated to 770 to 550 BC (4). The epigraphy of the Siloam Inscription places it slightly before or after 700 BC (5).

The end part of the water system was an artificial reservoir created by a monumental gravity dam that blocked the water flow from the Tyropoeon Valley into the Kidron Valley and from there to the Dead Sea. Large sections of the dam's upstream face have been exposed in recent

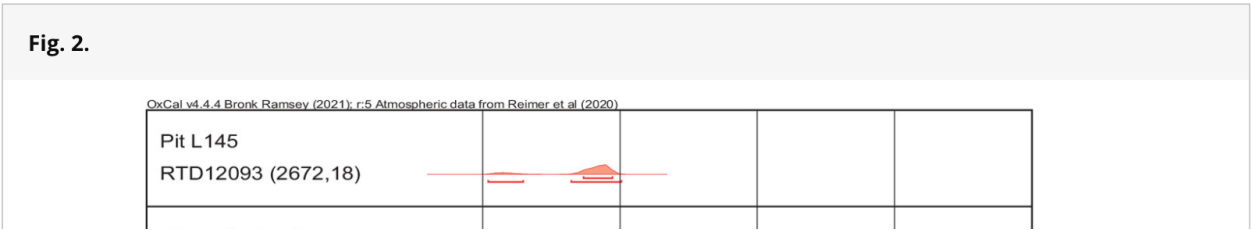
excavations (6, 7). It consists of both vertical and slanting sections, at least 19 m long, 11 m high, and 10 m wide (Fig. 1). A range of dates, from the 8th to the 1st centuries BC, was estimated over the years by numerous excavators who have exposed parts of the dam (7), including a single ¹⁴C date of the 4th to 3rd centuries BC (8).

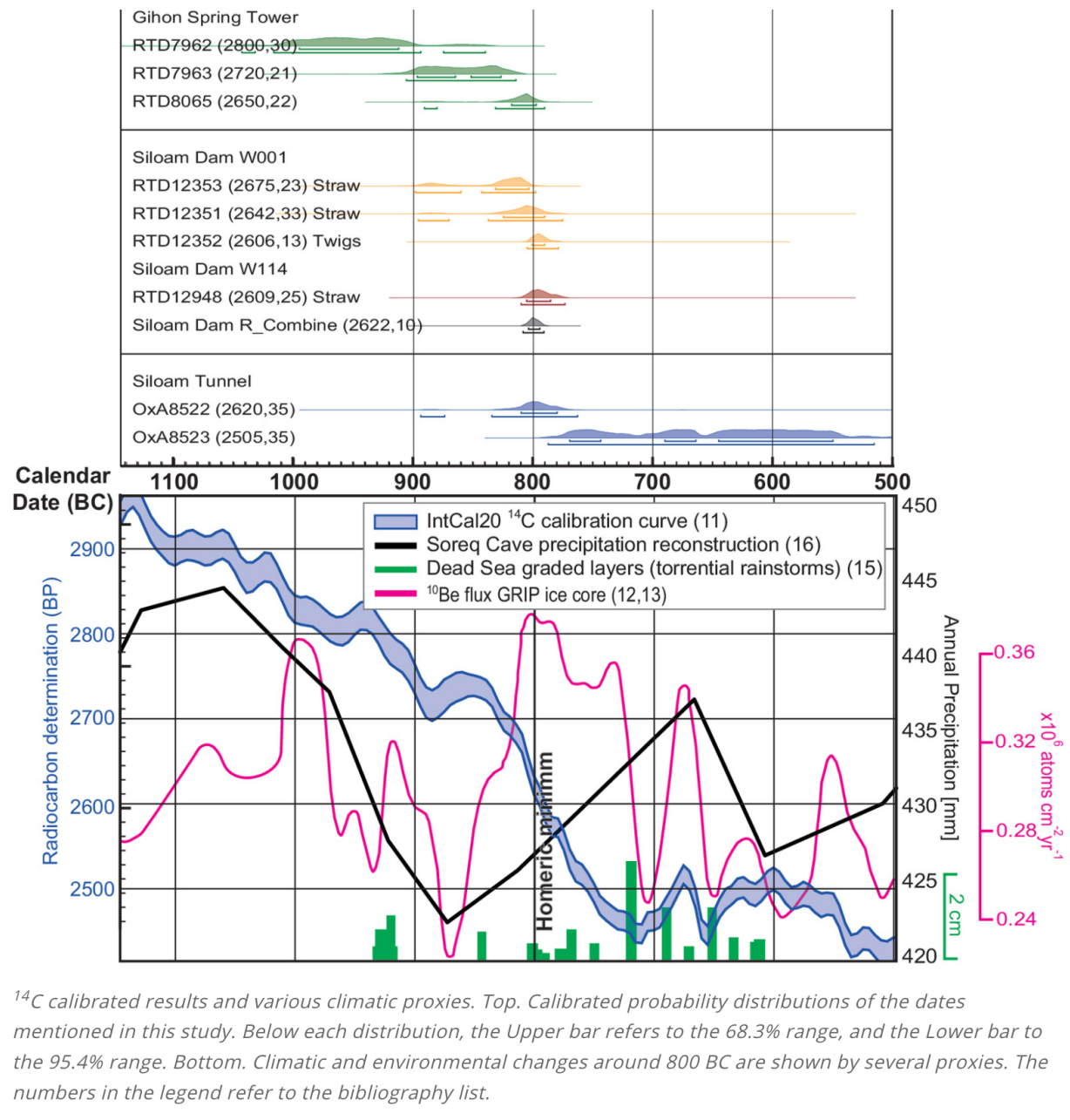
An additional installation in a domestic space is a large, oval-kidney-shaped Pit L145, measuring 14.9 × 14.4 × 6.3 m³. An early plaster layer identified in the pit was dated to 830 to 800 BC based on a charred inclusion resembling a small dung pellet. While Pit L145 clearly functioned as a cistern from the Late Roman Period onward, its use as a cistern in the 9th century BC is debatable (9).

Results and Discussion

Siloam Pool Dam.

We collected chunks of hard, yellowish mortar in between the outer stones of the dam’s vertical part (W001). Two separately collected uncharred straw samples were dated from the mortar (RTD 12353 and RTD 12351), and a third sample consisted of a few charred twigs (RTD 12352). From the slanting enforcement’s core (W114), we dated one sample of uncharred straw fragments from the mortar (RTD 12948). All four dates were very similar and could be averaged, giving a calibrated R _ combined date of 805 to 795 BC (68.3%) and 809 to 792 BC (95.4%) (Fig. 2). The combined date passes the X² test (T = 7.5, 5% = 7.8). The samples dating the Spring Tower and Pit L145 are slightly earlier and could reflect a *terminus post quem* for their construction. However, the uncharred straw dates of the Siloam Dam depict the actual building time, as long-term storage of fresh straw used for construction is unlikely.





The Early Water System.

The surprising dating results of the Siloam Pool Dam reveal that the Spring Tower (3) and the construction of the Siloam Pool reservoir were part of a single contemporary building project.

Channel II directed the water surplus from the Gihon Spring into the pool, which also collected rainwater running through the Tyropoeon Valley.

Pit L145, plastered around 830 to 800 BC, whether used for storage of water or other commodities, further attests to the city's changing needs, size, and nature at that time (9). While the precise dating of the Siloam Tunnel's construction remains obscure, one of the dates is also ~800 BC, but this sample can be too old due to an "old wood effect". The short-lived material measurement of the tunnel was $2,505 \pm 35$ BP. This ^{14}C level corresponds to the mid-8th century BC earthquake ($2,507 \pm 8$ BP) (10). However, this correlation can only be carefully suggested with the current data. The current data indicate that Channel II was initially the means through which the water reached the dammed reservoir in 800 BC, and the Siloam Tunnel replaced it several decades later.

The Climate Data of 850 to 800 BC.

The climate data provide a logical backdrop for the necessity of this renewed water system in Jerusalem around 800 BC. The Homeric grand solar minimum created an influx of radiocarbon, beginning around 830 BC, resulting in a steep slope in the radiocarbon calibration curve (11) and an increased Beryllium-10 production, measured in ice cores (12) and lake varves (13) (Fig. 2), simultaneously triggering a global climate change with significant variations in rain patterns around the year 800 BC (13). Valuable climate data near Jerusalem were derived from Dead Sea cores, as Kidron Valley is one of the riverbeds flowing eastward seasonally to the Dead Sea. The laminae sequence points significantly to a drier climate, represented by halite deposition beginning roughly at 850 BC (14), accompanied by torrential rainstorms causing flooding, as represented by graded layers (15). The Soreq Cave stalagmite data, located 20 km west of Jerusalem, suggest minimum precipitation around 850 BC, with a gradual increase in rainfall over the two following centuries (16) (Fig. 2). Reduced quantities of arboreal pollen can correlate with drier conditions (17), and $\Delta^{13}\text{C}$ values of olive pits also point to a change in rainfall pattern starting around 830 BC (18).

Several years of relatively dry winters with short episodes of heavy rainfall resulting in flash floods, perceived as unpredictable weather conditions, seem to have been the backdrop for the water system's building initiative. The ruling establishment of Judah, led in around 800 BC by King Jehoash or his successor Amaziah, provided a solution to the unpredictable weather conditions and water shortage by building the Siloam Pool dam, both for catching the rainwater and storing the water surplus brought by Channel II.

This study reveals the components of ancient waterworks in Jerusalem undertaken in 800 BC. It includes the water source fortification, the canal through which the water flowed, and the reservoir that collected it with the surface runoff. The construction was triggered by the period's drier climate, with sudden flash floods, undoubtedly also affected by the growing size of the city and pressure from the surrounding empires.

Materials and Methods

Using a binocular (Olympus SZ51), 3 to 5 mm fragments of uncharred straw and charred twigs were isolated from the bonding material (mortar). The samples were treated by acid–base–acid cleaning procedure, followed by graphitization and radiocarbon measurement at the Dangoor Research Accelerator Mass Spectrometer Laboratory ([10](#)). See the [SI Appendix](#) for complete details.

Data, Materials, and Software Availability

All study data are included in the article and/or [SI Appendix](#).

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Competing interests: The authors declare no competing interest.

Supporting Information

Appendix 01 (PDF)

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